



Evidence of the relationship between education expenditure by state and private sectors on solving resource curse in OPEC member countries

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ABSTRACT

This paper explores the link between education expenditure from both public and private sectors and the alleviation of the resource curse in 11 OPEC countries from 2000 to 2020. Utilizing the PMG-ARDL estimator, the study finds that a 1% increase in state education investment results in a 0.08% short-term and a 0.17% long-term rise in oil rent. In contrast, a 1% increase in private education investment leads to a 0.10% decrease in short-term and a 0.35% decrease in long-term oil rent. The research highlights the positive effects of GDP growth, trade volume, and electricity consumption in promoting economic diversification. The findings suggest that OPEC nations should enhance private sector education investments and focus on “greening” education to foster sustainable development and economic resilience.

1. introduction

Natural resources play a crucial role in fostering economic development and facilitating urban expansion. However, countries rich in resources often face the challenges of the resource curse, also known as the paradox of plenty (Yoshino et al., 2021; Wang et al., 2021; Ye and Rasoulinezhad, 2023; Zheng et al., 2023). This phenomenon arises due to factors such as excessive dependence on resource exports, fluctuating commodity prices, and inadequate institutional frameworks. The resource curse negatively impacts socioeconomic factors in these economies by increasing income inequality, obstructing diversification initiatives, and contributing to ongoing economic instability (Xu et al., 2023). Moreover, mismanagement and corruption associated with resource extraction can further weaken governance structures and erode public trust in institutions. Addressing these challenges requires effective governance, transparent management of resources, and strategic diversification to ensure sustainable and inclusive growth, thereby maximizing the benefits of natural resources for economic development and urban expansion (Rasoulinezhad and Taghizadeh-Hesary, 2022; Ding, 2023; Teng, 2023).

Education is crucial for improving social literacy regarding sustainable development, especially when tackling the challenges presented by the resource curse. Vargas-Merino et al. (2024) express that by educating citizens about sustainable practices, resource management, and the long-term consequences of resource extraction, societies can become more aware and proactive in mitigating the adverse impacts of resource dependency. Improved education fosters a better understanding of the importance of diversifying economies away from over-reliance on resource exports, promoting investment in human capital, innovation, and other sectors essential for long-term economic stability (Reimers, 2024). Moreover, educated populations are more likely to demand transparent governance, accountability, and responsible resource management practices from their governments, reducing the risk of corruption and mismanagement that often exacerbate the resource curse (Beata and Bartkus, 2023; Zeleniene and Pereira, 2021).

Education is essential for raising awareness of sustainable development and alleviating the resource curse, with both state and private sectors playing unique roles in these efforts. State-supported educational initiatives typically emphasize developing national curricula centered on sustainability principles, incorporating environmental education into

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formal education, and funding research on sustainable practices. These initiatives are vital for cultivating a comprehensive understanding of sustainability among the public and equipping future leaders and professionals to effectively address resource management challenges.

On the other hand, the private sector plays a significant role through corporate social responsibility (CSR) initiatives, partnerships with educational institutions, and investments in vocational training programs. Private sector involvement can bring practical insights and innovations into sustainability practices, promoting responsible resource management within industries. Furthermore, private sector initiatives can enhance the relevance of education by aligning educational programs with current market needs and technological advancements in sustainability. However, the impacts of state and private sector support for education on sustainability and resource curse mitigation can differ. State-funded education tends to prioritize broader societal goals and can ensure equitable access to education across all segments of society. In contrast, private sector initiatives may focus more on immediate business interests and profitability, potentially leading to disparities in educational quality and coverage. Balancing these interests requires collaboration and coordination between both sectors, leveraging their respective strengths to achieve comprehensive and effective education on sustainability and resource management.

In this paper, we aim to examine and compare the effects of state and private sector expenditures on education in relation to the resource curse in OPEC countries. OPEC is significant because its member nations are major oil producers, and their economies heavily rely on oil revenues. Understanding the role of education in mitigating the resource curse in these countries is essential for promoting sustainable development and economic diversification.

Education, supported by both state and private sectors, makes substantial contributions to addressing sustainability challenges and mitigating the resource curse. State-sponsored educational initiatives provide foundational knowledge and promote awareness of sustainable practices among the population. These initiatives often include integrating sustainability into national curricula, supporting research and development in sustainable technologies, and ensuring equitable access to education across diverse demographics. State involvement fosters a systemic approach to sustainability education, aiming to build a well-informed citizenry capable of advocating for and implementing sustainable policies. Conversely, the private sector contributes through innovative educational partnerships, corporate social responsibility initiatives, and investments in vocational training that focus on practical applications of sustainability principles. These efforts bring industry-specific expertise, technological advancements, and real-world examples of sustainable practices into educational settings. Private sector initiatives often emphasize skill development and job readiness in sustainable industries, thereby enhancing the relevance and effectiveness of sustainability education in preparing future professionals.

The research structure outlined begins with a comprehensive literature review and gap identification (Section 1), which lays the foundation by examining existing knowledge and identifying areas where further research is needed. Section 2 provides a background discussion, building upon the literature review to contextualize the research within current theories and debates related to sustainability and the resource curse. Section 3 delves deeper into the research methodology and variables used, outlining how data was collected, analyzed, and interpreted to address the research questions or hypotheses. Section 4 presents the findings of the study, offering empirical evidence and insights gained from the research process. This section typically includes data analysis, results, and discussion of key findings in relation to the research objectives. Finally, Section 5 concludes the research with a summary of findings, implications for theory and practice, and policy recommendations. This section connects the research outcomes to broader implications for sustainable development and strategies to mitigate the resource curse, providing actionable recommendations for policy-makers, stakeholders, and practitioners.

2. Literature review

The literature on resource consumption, economic growth, development, and industrialization highlights the intricate relationship between natural resource availability and economic outcomes. Historically, economic growth and industrialization have depended significantly on the extraction of natural resources, including minerals, fossil fuels, and agricultural products, which have fueled production and infrastructure development (Kerner et al., 2023; Singh et al., 2024). This resource-driven model has enabled rapid economic expansion in various countries by supplying crucial inputs for industrial production and export-led growth strategies (Gonzalez-Val and Pueyo, 2019; Rasoulinezhad, 2020). However, the sustainability of these growth patterns is increasingly challenged by concerns regarding environmental degradation, resource depletion, and the effects of climate change (Khan et al., 2022). Additionally, the literature emphasizes the unequal distribution of benefits from resource-intensive growth, which often worsens income inequality and social disparities (Jaimes and Gerlagh, 2020; Lee and He, 2022).

The phenomenon of the Resource Curse has been extensively studied in the context of OPEC (Organization of the Petroleum Exporting Countries) member states, where the paradox of abundant natural resource wealth coupled with economic underperformance is well-documented. Countries such as Nigeria, Venezuela, and Iraq illustrate the difficulties associated with a heavy dependence on oil revenues, including economic volatility, governance issues, and insufficient economic diversification (Ozcan et al., 2023; Barczikay et al., 2020). The resource curse in OPEC nations is often linked to factors like rent-seeking behavior, weak institutions, and ineffective resource management policies (Asif et al., 2020; Lacarcel et al., 2024). Despite generating significant oil revenues, these countries frequently face socio-economic inequalities, environmental degradation, and political instability, which impede sustainable development efforts (Mlambo, 2022).

Education plays a pivotal role in influencing resource consumption patterns and promoting sustainable practices across societies. Research suggests that higher levels of education are associated with greater awareness of environmental issues, leading to more conscientious consumption habits and reduced environmental impact (Hu and Zheng, 2023; Xio et al., 2024). Educated individuals tend to make more informed decisions regarding energy use, waste management, and resource allocation, contributing to overall resource efficiency and conservation efforts (Yin et al., 2024). Moreover, education fosters innovation and technological advancements that can enhance resource efficiency in various sectors, from agriculture to manufacturing (Chen et al., 2023; Huang et al., 2023).

Sustainable education plays a crucial role in fostering social sustainable literacy, which encompasses understanding, skills, and values necessary for individuals and communities to engage effectively in sustainable practices and contribute to societal well-being. Sustainable education integrates environmental, social, and economic dimensions of sustainability into curricula and learning experiences, equipping learners with knowledge about the interconnectedness of global challenges such as climate change, biodiversity loss, and social inequality (Mora et al., 2020; Vargas-Merino et al., 2024). By promoting critical thinking, ethical reasoning, and systems-level understanding, sustainable education empowers individuals to make informed decisions that support sustainable development goals and address pressing environmental and social issues (Figueiro et al., 2022). Moreover, it encourages active citizenship and collective action towards building resilient, inclusive, and equitable societies, where the needs of current and future generations are met without compromising the ability of ecosystems to thrive (Nousheen et al., 2020; Hermann and Bossle, 2020).

The existing literature on the resource curse primarily emphasizes macroeconomic indicators and governance structures when analyzing its effects on economies, especially in oil-rich countries. However, there is a notable gap in understanding how education can serve as a

mitigating factor in the resource curse. While some studies acknowledge education's indirect impact through governance and human capital development, few thoroughly investigate how educational policies and literacy directly influence resource management and sustainability outcomes. This paper aims to fill this gap by exploring the specific mechanisms through which education enhances social sustainability literacy and helps alleviate the resource curse. In contrast to previous research that mainly focuses on economic impacts, this study will examine educational strategies and their implications for sustainable development, providing a more nuanced understanding of how improving educational outcomes can foster more resilient and diversified economies in resource-dependent countries.

3. Background discussion

OPEC countries, endowed with vast reserves of non-renewable resources, predominantly oil, face significant challenges related to their extraction and utilization. As OPEC remains optimistic about future oil demand, projecting a 12% increase by 2030 compared to 2021, these projections underscore the continued reliance on fossil fuels despite global calls for decarbonization (OPEC, 2023a). With approximately 79.5% of the world's proven oil reserves concentrated within OPEC member countries, the region holds immense strategic importance in global energy markets (OPEC, 2023b). However, the overuse of these non-renewable resources poses environmental, economic, and social risks. Environmental concerns include greenhouse gas emissions, biodiversity loss, and pollution from extraction and refining processes. Economically, over-reliance on oil revenues can lead to volatility in national budgets and hinder diversification efforts, while socially, it can exacerbate inequalities and dependency on fluctuating global markets.

Education serves as a crucial transmission channel in mitigating the resource curse by influencing various facets of societal and economic development.

State-funded education expenditures usually focus on broad accessibility and standardized curricula designed to promote civic literacy, critical thinking, and technical skills essential for sustainable resource management. Governments invest in educational initiatives that not only seek to improve overall educational attainment but also emphasize the development of competencies needed for effective resource stewardship (Hu and Zheng, 2023; Zhou et al., 2024; Li et al., 2024). These initiatives integrate environmental sustainability and responsible governance practices into educational frameworks, preparing future generations to tackle complex environmental challenges and engage actively in sustainable development efforts (Yazdani et al., 2023). By incorporating these principles into national curricula and policies, states aim to cultivate an informed and engaged citizenry that can contribute to both economic growth and environmental conservation.

In contrast, private education expenditures, driven by corporate social responsibility or market interests, tend to focus on providing specialized training aligned with industry needs. This emphasis enhances technical skills and fosters innovation within resource sectors, aiming to fulfill specific workforce requirements and stimulate economic growth. Private sector involvement may utilize industry expertise to tailor educational programs that address skills gaps and technological advancements in resource extraction and management. However, the profit-oriented nature of private education can sometimes prioritize short-term economic benefits over broader societal and environmental considerations. This could result in inadequate attention to sustainable practices, environmental stewardship, and ethical issues in resource extraction processes (Zhou et al., 2024; Zhang et al., 2022; Liang et al., 2024). Therefore, it is essential to balance private sector initiatives with regulatory frameworks and collaborative efforts with state-funded educational institutions to ensure that educational investments positively impact sustainable development goals and address the long-term challenges of resource dependency.

4. Variables and method of estimation

The main idea of this paper is to provide evidence of the relationship between education expenditure by both state and private sectors and the mitigation of the resource curse in 11 OPEC member countries, including Qatar, Indonesia, the United Arab Emirates, Algeria, Nigeria, Ecuador, Angola, Iran, Kuwait, Saudi Arabia, and Congo. These countries were selected based on data availability and relative geopolitical stability, while other OPEC countries were excluded due to these constraints. The research spans from 2000 to 2020, examining the resource curse, proxied by oil rent, as the dependent variable. The independent variables include state investment in education, private investment in education, gross domestic production, trade volume, and electricity consumption. Each independent variable is essential in understanding its impact on the resource curse. State and private investments in education can enhance human capital, reducing reliance on oil rents and fostering economic diversification. Gross domestic production reflects the overall economic health, influencing the capacity to invest in various sectors. Trade volume indicates economic openness and integration into the global economy, which can mitigate the adverse effects of the resource curse. Electricity consumption serves as a proxy for infrastructure development and economic activity, further influencing the resource curse. Below is Table 1 that reports the variables, their types, abbreviations, and sources of data collection for the study.

State and private investments in education, higher GDP, increased trade volume, and higher electricity consumption can collectively mitigate the resource curse by fostering economic diversification and human capital development. Improved educational outcomes create a skilled workforce capable of engaging in a wide range of economic activities beyond oil extraction. Education equips individuals with the knowledge, skills, and competencies needed to innovate, adapt to new technologies, and contribute to various industries such as manufacturing, services, and technology. As a result, a well-educated population can drive entrepreneurial ventures, enhance productivity across different sectors, and attract foreign investments. This diversification reduces the economy's vulnerability to fluctuations in oil prices and promotes more stable and inclusive economic growth. Additionally, education fosters critical thinking and civic engagement, which can improve governance and the effective management of natural resources. A higher GDP indicates a more diverse and resilient economy because it often reflects the development of multiple economic sectors rather than an overreliance on a single resource such as oil. This diversity allows the economy to better withstand shocks, such as fluctuations in commodity prices, and fosters overall stability and growth. Greater trade volume reflects economic openness by showing that the country is actively participating in global markets. This openness enhances competitiveness as domestic firms must meet international standards and innovate to succeed. Additionally, increased trade volume provides access to a broader range of goods, services, and technologies, which can drive

Table 1
Specification of the variables and data.

Variable	Type	Abbreviation	Source of data collection
Resource curse (oil rent)	Dependent variable	OR	World Bank
State investment in education	Independent variable	SIE	UNESCO Institute Statistics, Local Databases
Private investment in Education	Independent variable	PIE	UNESCO Institute Statistics, Local Databases
Gross domestic product	Independent variable	GDP	World Bank
Trade volume	Independent variable	TV	UN Comtrade
Electricity consumption	Independent variable	EC	International Energy Agency (IEA)

further economic growth and diversification. It also allows the country to establish advantageous trade relationships, attract foreign investment, and integrate more fully into the global economy, all of which help build a stronger and more resilient economic framework. Higher electricity consumption reflects industrialization and urbanization, indicating the growth of energy-intensive industries and the development of urban areas where economic activities are concentrated. Industrialization involves the development and expansion of manufacturing and production facilities, which require substantial electricity to operate machinery, maintain production lines, and support technological advancements. This shift towards a more industrialized economy diversifies economic activities beyond the extraction of natural resources, creating jobs, fostering innovation, and increasing productivity. Similarly, urbanization entails the growth of cities and towns, which drives demand for electricity to power homes, businesses, infrastructure, and services. As more people move to urban areas, the concentration of economic activities accelerates, leading to the establishment of new businesses, enhanced service sectors, and improved infrastructure. This, in turn, supports a wide range of economic activities, from retail and hospitality to finance and healthcare, further reducing reliance on the oil sector and contributing to a more balanced and resilient economy. Table 2 outlines the expected signs of the coefficients.

The estimation of the coefficients is based on a panel data analysis of data for OPEC countries. This analysis ensures robust results by considering both cross-sectional and time-series dimensions. To start, the panel data should be checked for slope homogeneity, for which we utilized Swamy (1970)'s approach. This method tests whether the relationships between variables are consistent across different countries in the panel. Following this, the presence of cross-sectional dependency is evaluated using four tests, ensuring that the interdependencies between countries are properly accounted for in the analysis. Next, the stationarity of the panel data is assessed using the Bai and Ng (2004) panel stationary test. This test determines whether the data series are stable over time, which is crucial for reliable econometric modeling. Once stationarity is confirmed, the cointegration relationships between the variables are explored using the Westerlund (2007) technique. Cointegration analysis determines whether a long-term equilibrium relationship exists among the variables, despite short-term fluctuations. The study employs the Pooled Mean Group (PMG) - Autoregressive Distributed Lag (ARDL) model as its estimator. This PMG-ARDL approach enables the estimation of both short-term dynamics and long-term relationships within a panel data framework, allowing for variations in short-term adjustments while maintaining a common long-term relationship across countries. This thorough methodology guarantees that the estimated coefficients are robust, effectively capturing the intricate interactions between the variables in the context of OPEC countries.

Table 2
Anticipation of the impacts of the variables.

Factor	Expected sign	Reason
State Investment in Education	Negative	Enhances human capital, leading to economic diversification and reduced dependency on oil rents.
Private investment in Education	Negative	Complements state efforts, improves educational quality, fosters innovation, and promotes private sector growth.
Gross domestic product	Negative	Reflects a diverse and resilient economy, reducing the relative importance of oil rents.
Trade volume	Negative	Indicates economic openness, enhancing competitiveness, access to international markets, and economic diversification.
Electricity consumption	Negative	Signifies industrialization and urbanization, supporting broader economic activities and reducing reliance on the oil sector.

5. Results

Table 3 presents the results of the Swamy (1970) test, which reject the null hypothesis. This outcome implies that the relationships between the variables vary across different countries in the panel, indicating slope heterogeneity.

Next, cross-sectional dependency is evaluated for the panel of OPEC countries. Table 4 lists the findings from four different tests, revealing significant interconnections among the OPEC members.

Following this, the Bai and Ng (2004) test is conducted to check the stationarity level of the series. Based on the test findings in Table 5, it can be noted that the variables exhibit stationarity at levels 0 and 1.

The Westerlund technique is used to evaluate the long-term association between the independent variables. Table 6 lists the results and confirms that the independent variables exhibit cointegrated features.

The coefficients are estimated using the PMG-ARDL method. Table 7 presents the results as follows.

The estimation results indicate that a 1% increase in state investment in education leads to a short-term increase of 0.08% and a long-term increase of 0.17% in oil rent (resource curse) among OPEC countries. This suggests that higher state investment in education initially stimulates economic diversification and human capital development, which can reduce dependence on oil revenues in the short term. However, over the long term, the impact on oil rent becomes more pronounced, potentially reflecting complex interactions between educational development, economic structure, and resource dependency.

The analysis reveals that a 1% increase in private investment in education corresponds to a reduction of 0.10% in short-term and 0.35% in long-term oil rent among OPEC countries. This finding suggests that increased private investment in education fosters a more diversified economy by enhancing educational quality, promoting innovation, and expanding opportunities beyond the oil sector. In the short term, private investment in education stimulates immediate improvements in human capital and economic efficiency, reducing the immediate reliance on oil rents. Over the long term, as the educational sector matures and produces a highly skilled workforce, it contributes significantly to economic diversification.

The findings confirm the positive role of GDP expansion in mitigating the resource curse phenomenon among OPEC countries. An increase in GDP indicates a more diversified and resilient economy, less reliant on oil rents. As GDP expands, it signifies growth in sectors beyond oil extraction, such as manufacturing, services, and technology. This economic diversification reduces the economy's vulnerability to fluctuations in oil prices and enhances overall stability. Moreover, a higher GDP fosters higher levels of investment in infrastructure, education, and social programs, which further contribute to sustainable development and diminishes the negative impacts traditionally associated with oil dependency. Therefore, the positive correlation between GDP expansion and mitigating the resource curse underscores the importance of economic diversification strategies in ensuring long-term prosperity and stability in OPEC economies.

The coefficients of trade volume and electricity consumption show positive effects in both the short and long terms. This indicates that increasing trade volume and electricity consumption contribute to mitigating the resource curse phenomenon in OPEC countries for several reasons. In the short term, higher trade volume enhances economic openness and diversification. OPEC countries that engage more extensively in international trade can reduce their dependency on oil exports by accessing new markets and diversifying their export base. This

Table 3
Slope homogeneity of Swamy (1930).

Statistic	Test value	Prob.
Swamy stat.	18.943	0.000
Adjusted Swamy stat.	24.121	0.000

Table 4

Cross-sectional dependency test.

Test	Value	Prob.
Breusch-Pagan LM	742.311	0.000
Pesaran Scaled LM	21.437	0.000
Bias-corrected Scaled LM	28.693	0.000
Pesaran CD test	6.573	0.000

Table 5

Bai and Ng (2004) examination.

Factor	I (0)	I (1)
Resource curse (oil rent)	−0.275 (0.795)	−3.276 (0.000)
State investment in education	−0.561 (0.942)	−4.005 (0.000)
Private investment in Education	−4.371 (0.005)	−16.109 (0.000)
Gross domestic product	−1.664 (0.857)	−5.411 (0.000)
Trade volume	−5.796 (0.053)	−17.312 (0.000)
Electricity consumption	−0.593 (0.712)	−3.697 (0.000)

Table 6

Cointegration test.

Statistic	Value	Z-value	P-value
G_τ	−6.518	−6.559	0.000
G_a	−20.896	−3.923	0.000
P_τ	−8.793	−4.379	0.035
P_a	−26.592	−6.744	0.001

Table 7

Pooled Mean Group- ARDL estimation result.

Perspective	Factor	Coefficient	Prob.
Long-term	D (Resource curse (oil rent))	0.637	0.339
	State investment in education	0.176	0.035
	Private investment in Education	−0.351	0.047
	D (Private investment in Education)	−0.046	0.007
	Gross domestic product	0.634	0.089
	Trade volume	0.516	0.064
	Electricity consumption	0.149	0.050
Short-term perspective	State investment in education	0.086	0.066
	Private investment in Education	−0.105	0.092
	Gross domestic product	0.436	0.007
	Trade volume	0.053	0.003
	Electricity consumption	0.097	0.018
	ECT (−1)	−0.789	0.033

reduces vulnerability to fluctuations in oil prices and strengthens economic resilience. In the long term, sustained trade volume supports the development of a more diversified economy. Increased exposure to international markets encourages technological advancements, adoption of best practices, and enhances competitiveness. This leads to the growth of non-oil sectors such as manufacturing, services, and agriculture, thereby reducing reliance on oil rents as the primary source of revenue. Regarding electricity consumption, in the short term, increased electricity consumption signifies industrialization and urbanization. This supports broader economic activities beyond oil extraction, such as manufacturing and construction. Industrialization driven by higher electricity consumption creates job opportunities, boosts productivity, and generates income diversification.

In the long term, continuous growth in electricity consumption promotes ongoing industrial development and economic diversification. It enables the expansion of urban areas and the improvement of infrastructure, both of which are vital for economic growth in non-oil sectors. Furthermore, reliable and accessible electricity is essential for attracting foreign investment and fostering the growth of industries that can thrive independently of oil revenues.

Continuing, diagnostic tests are conducted to ensure the robustness and reliability of the findings. Table 8 reports the results of these tests, affirming that the employed model is credible. These diagnostic tests assess various aspects such as multicollinearity, heteroscedasticity, autocorrelation, and specification errors.

After that, the robustness of the findings is checked by changing the indicator of the resource curse from oil rent to coal rent. Table 9 presents the results, demonstrating that the findings remain consistent and suitable for interpretation. This robustness check confirms that the relationships identified in the study are not specific to oil rent alone but are also applicable when using coal rent as an indicator.

6. Conclusion and policy recommendations

This paper presents strong evidence of the relationship between education expenditure from both state and private sectors and the mitigation of the resource curse in 11 OPEC member countries: Qatar, Indonesia, the United Arab Emirates, Algeria, Nigeria, Ecuador, Angola, Iran, Kuwait, Saudi Arabia, and Congo. These countries were chosen based on data availability and relative geopolitical stability, while other OPEC nations were excluded due to these limitations. Covering the period from 2000 to 2020, the research indicates that a 1% increase in state education investment leads to a short-term increase of 0.08% and a long-term increase of 0.17% in oil rent. This suggests that state investment in education initially stimulates economic diversification and human capital development, helping to reduce dependence on oil revenues in the short term. However, the long-term impact on oil rent is more significant, potentially reflecting complex interactions among educational development, economic structure, and resource dependency. Additionally, a 1% increase in private investment in education results in a 0.10% reduction in short-term oil rent and a 0.35% reduction in the long term. This indicates that private investment in education promotes a more diversified economy by enhancing educational quality, fostering innovation, and expanding opportunities beyond the oil sector. Such investment leads to immediate improvements in human capital and economic efficiency, decreasing reliance on oil rents in the short term, while a highly skilled workforce significantly aids economic diversification in the long term. The findings also highlight the positive role of GDP expansion in mitigating the resource curse among OPEC countries. An increase in GDP signifies a more diversified and resilient economy that is less reliant on oil revenues. As GDP grows, it indicates development in sectors beyond oil extraction, such as manufacturing, services, and technology, thus reducing vulnerability to fluctuations in oil prices and enhancing overall stability. Furthermore, higher GDP encourages greater investment in infrastructure, education, and social programs, supporting sustainable development and alleviating the negative effects associated with oil dependency. The positive correlation between GDP expansion and the mitigation of the resource curse emphasizes the importance of economic diversification strategies for ensuring long-term prosperity and stability in OPEC economies. The coefficients for trade volume and electricity consumption also show positive effects in both the short and long term, suggesting that increased trade volume and electricity consumption contribute to alleviating the resource curse. Higher trade volume promotes economic openness and diversification, reducing reliance on oil exports by accessing new markets and broadening the export base, thus strengthening economic resilience. Over the long term, sustained trade volume

Table 8

Diagnostic tests.

Test	LM value	Prob.
Serial correlation	0.074	0.173
Breusch-Pagan- Godfrey	0.564	0.259
Jarque-Bera	0.842	0.669
Ramsey RESET test	1.647	0.733

Table 9
Re-estimation by new dependent variable (Coal rent).

Perspective	Factor	Coefficient	Prob.
Long-term	D (Resource curse (coal rent))	0.536	0.077
	State investment in education	0.274	0.013
	Private investment in Education	−0.200	0.073
	D (Private investment in Education)	−0.017	0.033
	Gross domestic product	0.476	0.003
	Trade volume	0.425	0.027
	Electricity consumption	0.200	0.002
Short-term perspective	State investment in education	0.163	0.073
	Private investment in Education	−0.264	0.036
	Gross domestic product	0.221	0.084
	Trade volume	0.033	0.095
	Electricity consumption	0.162	0.057
	ECT (−1)	−0.722	0.003

fosters the development of a more diversified economy, encouraging technological advancements, best practices, and enhanced competitiveness, leading to growth in non-oil sectors and decreasing reliance on oil rents. Regarding electricity consumption, higher consumption levels reflect industrialization and urbanization in the short term, supporting broader economic activities beyond oil extraction, including manufacturing and construction. This industrialization, driven by increased electricity consumption, generates job opportunities, boosts productivity, and fosters income diversification. Over the long term, sustained growth in electricity consumption supports ongoing industrial development and economic diversification, facilitating urban expansion and infrastructure development.

Based on the findings of this paper, OPEC countries should implement practical policies to mitigate the resource curse by prioritizing education investment from both the private and public sectors. Governments can stimulate private sector investment in education through tax incentives, subsidies, and public-private partnerships, which would enhance educational quality and encourage innovation. These efforts will contribute to economic diversification and reduce dependence on oil revenues. Additionally, state investments should focus on “greening” education by integrating sustainability and environmental studies into curricula, thereby preparing a workforce skilled in green technologies and sustainable practices. This combined strategy—enhancing private sector involvement in education while allocating state resources toward environmentally-focused education—can drive economic diversification, promote sustainable development, and mitigate the long-term effects of the resource curse. Furthermore, policies should support infrastructure development and ensure a reliable electricity supply to facilitate industrialization and urbanization, ultimately creating a more resilient economic environment that is less reliant on oil revenues.

For future studies, it is recommended to explore the impact of green education plans on mitigating the resource curse in OPEC countries. Researchers should investigate how integrating sustainability and environmental education into the curriculum can develop a workforce skilled in green technologies and sustainable practices, ultimately diversifying the economy and reducing reliance on oil rents. Additionally, examining the relationship between green taxes and the resource curse in OPEC countries is crucial. Future studies should analyze how implementing green taxes on carbon emissions and other environmental pollutants can incentivize a shift towards sustainable industries, further promoting economic diversification.

CRedit authorship contribution statement

Yu Li: Writing – original draft, Project administration, Methodology, Investigation, Conceptualization. **Wei Wei:** Validation, Formal analysis, Data curation.

Declaration of competing interest

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all the authors. We declare that there is no conflict of interest.

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Data availability

Data will be made available on request.

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